

Module 1: Introduction to Mobile Computing

1. Describe signal propagation and explain path loss effect.
2. Explain cellular system.
3. Compare DSSS and FHSS.
4. Define spread spectrum and explain any one method used for spread spectrum.
5. Explain frequency reuse used in cellular communication.
6. Explain any two applications of electromagnetic spectrum in detail.
7. Explain ranges of signal propagation i.e. transmission detection with diagram.
8. Why the size of the cell is kept smaller in cellular network? Explain.
9. Explain the components of mobile computing.
10. Explain various types of antennas along with their radiation pattern.

Module 2: GSM Mobile Services

1. Explain UTRAN architecture.
2. Differentiate between 2G and 3G.
3. List various security services offered by GSM. Explain A3, A5, A8 algorithms briefly.
4. Differentiate between GSM and GPRS.
5. Explain different databases used in GSM system.
6. Explain any two radio interfaces used in GSM system.
7. Explain GPRS architecture.

Module 3: Mobile Networking

1. Justify the role of Home Agent and Foreign Agent in Mobile IP network.
2. Describe Mobile IP network.
3. Explain Mobile IP topology in detail.